DAY-12 – TUPLES IN PYTHON

Python 90 Days Challenge • Compact GitHub Notes

1. What is a Tuple?

A tuple is a built-in data structure in Python used to store multiple items in a single variable. Tuples are ordered, immutable, and allow duplicate values.

```
fruits = ("Apple", "Banana", "Mango")
print(fruits)
```

Output:

```
('Apple', 'Banana', 'Mango')
```

2. Features of Tuples

• Ordered: Items maintain their order.
• Immutable: Items cannot be changed after creation.
• Duplicates Allowed: A tuple can contain duplicate values.
• Multiple Data Types: A tuple can store different data types.
• Indexed: Items can be accessed using their index.

3. Creating a Tuple

```
# Empty tuple
t1 = ()
# Tuple with integers
t2 = (10, 20, 30)
# Different data types
t3 = (10, "Python", 3.5, True)
# Without parentheses
t4 = 10, 20, 30
# Single-item tuple
t5 = (10,)
print(t2)
print(t5)
```

Output:

```
(10, 20, 30)
(10,)
```

4. Accessing Tuple Elements

```
numbers = (10, 20, 30, 40, 50)
print(numbers[0])
print(numbers[2])
print(numbers[-1])
```

Output:

```
10
30
50
```

5. Tuple Slicing

Syntax: **tuple[start:stop:step]**

```
numbers = (10, 20, 30, 40, 50)
print(numbers[1:4])
print(numbers[:3])
print(numbers[::2])
```

Output:

```
(20, 30, 40)
(10, 20, 30)
(10, 30, 50)
```

6. Tuple Methods

Method	Description	Example
`count()`	Returns the number of times an element appears.	`t.count(10)`
`index()`	Returns the index of the first occurrence of an element.	`t.index(20)`

```
numbers = (10, 20, 10, 30, 10)
print(numbers.count(10))
print(numbers.index(20))
```

Output:

```
3
```

```
1
```

7. Tuple Operations

```
# Concatenation
print((1, 2) + (3, 4))
# Repetition
print((1, 2) * 3)
# Membership
print(20 in (10, 20, 30))
print(40 not in (10, 20, 30))
# Length
print(len((10, 20, 30)))
```

Output:

```
(1, 2, 3, 4)
(1, 2, 1, 2, 1, 2)
True
True
3
```

8. Tuple Immutability

Tuple elements cannot be modified after creation. To change values, convert the tuple to a list, modify it, and convert it back.

```
numbers = (10, 20, 30)
temp = list(numbers)
temp[0] = 100
numbers = tuple(temp)
print(numbers)
```

Output:

```
(100, 20, 30)
```

9. Tuple Packing and Unpacking

```
student = ("Payal", 21, "BCA")
name, age, course = student
print(name)
print(age)
print(course)
```

Output:

```
Payal
21
BCA
```

10. Nested Tuples

```
data = ((1, 2), (3, 4), (5, 6))
print(data[0])
print(data[1][1])
```

Output:

```
(1, 2)
4
```

11. Difference Between List and Tuple

List	Tuple
Created using `[]`	Created using `()`
Mutable	Immutable
Many modification methods	Two built-in methods: `count()` and `index()`
Suitable for changeable data	Suitable for data that should remain unchanged

12. Real-World Applications

- Storing fixed student details.
- Representing coordinates such as (x, y).
- Storing RGB color values.
- Returning multiple values from a function.

13. Practice Questions

1. Create a tuple containing five numbers.
2. Access the first and last elements.
3. Print the first three elements using slicing.
4. Count how many times a number appears.
5. Find an element's index using `index()`.
6. Concatenate two tuples.
7. Repeat a tuple three times.
8. Convert a tuple into a list and modify an element.
9. Demonstrate packing and unpacking.
10. Create a nested tuple and access an inner element.

14. Interview Questions

Q1. What is a tuple? An ordered and immutable collection of elements.
Q2. Is a tuple mutable? No, it is immutable.
Q3. Can tuples contain duplicates? Yes.
Q4. How do you create a single-element tuple? Use a trailing comma, e.g. `(10,)`.
Q5. How many built-in tuple methods are there? Two: `count()` and `index()`.
Q6. Can a tuple contain different data types? Yes.
Q7. What is tuple unpacking? Assigning tuple elements to separate variables.
Q8. Why are tuples useful? For ordered data that should not be modified.

Day-12 Completed: Tuples in Python